

A weak- L^4 endpoint obstruction for empirical covariance

Candidate counterexample packet for arXiv:1509.02322

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Status. Candidate counterexample, likely valid. The result gives a complete negative answer to the natural square-root endpoint formulation under the source’s weak marginal-tail hypothesis. It does not rule out a logarithmically weakened endpoint estimate or address a uniform finite-fourth-moment theorem.

1 Source question and precise scope

Guedon, Litvak, Pajor, and Tomczak-Jaegermann study the empirical covariance fluctuation

$$S = \sup_{a \in S^{n-1}} \left| \frac{1}{N} \sum_{i=1}^N (\langle X_i, a \rangle^2 - \mathbb{E} \langle X_i, a \rangle^2) \right|$$

under the marginal condition

$$\mathbf{H}(\phi) : \quad \mathbb{P}\{|\langle X_i, a \rangle| \geq t\} \leq \tau/\phi(t) \quad (a \in S^{n-1}, t > 0).$$

The source explicitly says that the $p = 4$ endpoint is unknown (page 6) and asks whether its $4 < p \leq 8$ remainder can be improved to exponent $1/2$ (page 20) [1]. The two source passages are reproduced below.

bounds for S involving the term M^2/n like the bound (5) are of interest only for $p \geq 4$. We don’t know if it holds in the case $p = 4$.

The main novelty of our proof is a delicate analysis of the behavior of norms of submatrices, namely quantities A_k and B_k , $k \leq N$, defined in (6) below. This analysis is done in Theorem 2.1 which is in the heart of the technical part of the paper and it will be presented in the next section. The estimates for B_k are responsible for RIP, Theorem 1.1 while the estimates for A_k are responsible for KLS problem, Theorem 1.2

As usual in this paper $C, C_0, C_1, \dots, c, c_0, c_1, \dots$ always denote absolute positive constants whose values may vary from line to line.

Figure 1: Source page 6: the endpoint sentence.

The precise statement tested here is the natural endpoint extrapolation

$$S \leq C \left(\frac{1}{N} \max_{i \leq N} |X_i|^2 + \sqrt{\frac{n}{N}} \right) \tag{1}$$

with a universal constant and probability tending to one, assuming $\mathbf{H}(t^4)$ with a fixed tail parameter. We disprove (1) already for centered isotropic scalar variables.

Z , the hypothesis $H(\phi)$, $\phi(t) = t^2$ is satisfied but $N \sim n \log n$ is needed for the covariance matrix to be well approximated by the empirical covariance matrices with probability $1/2$.

We also would like to mention that we don't know how sharp the power γ/p appearing in the bound below is. In particular, it is not clear if it can be improved to $1/2$.

Theorem 4.5 *Let $4 < p \leq 8$ and $\phi(t) = t^p$. Let $X_1, \dots, X_N$ be independent random vectors in $\mathbb{R}^n$ satisfying hypothesis $\mathbf{H}(\phi)$. Let $\varepsilon \leq \min\{1, (p-4)/4\}$ and $\gamma = p-4-2\varepsilon$. Then with probability larger than*

$$1 - 8e^{-n} - 2\varepsilon^{-p/2} \max\{N^{-3/2}, n^{-(p/4-1)}\}$$

one has

$$\sup_{a \in S^{n-1}} \left| \frac{1}{N} \sum_{i=1}^N (\langle X_i, a \rangle^2 - \mathbb{E} \langle X_i, a \rangle^2) \right| \leq C \left(\frac{1}{N} \max_{i \leq N} |X_i|^2 + C(p, \varepsilon) \left(\frac{n}{N} \right)^{\gamma/p} \right),$$

where

$$C(p, \varepsilon) = (p-4)^{-1/2} \varepsilon^{-4(2+\varepsilon)/p}.$$

Figure 2: Source page 20: the square-root-rate question and Theorem 4.5 bound.

2 Proof intuition

At the weak- L^4 endpoint, the square X^2 is only weak- L^2 . A Pareto tail of exponent two has a logarithmically divergent second moment. Truncate it at $\sqrt{N}$: this makes every sample satisfy the maximum-column correction in (1), but retains variance of order $\log N$. The empirical second moment consequently fluctuates on the larger scale $\sqrt{\log N/N}$. A fourth-moment computation and Paley–Zygmund make this mechanism quantitative with constant probability.

3 Counterexample theorem

Theorem 1. *For every integer N with $\log N \geq 6$, there is a centered isotropic real random variable $X = X^{(N)}$ such that*

$$\mathbb{P}\{|X| \geq t\} \leq t^{-4} \quad (t > 0).$$

For independent copies $X_1, \dots, X_N$,

$$\mathbb{P} \left\{ \left| \frac{1}{N} \sum_{i=1}^N (X_i^2 - 1) \right| \geq \frac{1}{4} \sqrt{\frac{\log N}{N}} \right\} \geq \frac{1}{76},$$

whereas, deterministically,

$$\frac{1}{N} \max_{i \leq N} X_i^2 \leq N^{-1/2}.$$

Proof. Put $L = \sqrt{N}$. Let Z be the nonnegative random variable with survival function

$$\mathbb{P}\{Z \geq u\} = \begin{cases} 1, & 0 \leq u \leq 1, \\ u^{-2}, & 1 < u \leq L, \\ 0, & u > L. \end{cases}$$

Thus Z has density $2u^{-3}$ on $[1, L)$ and an atom of mass L^{-2} at L . Write $\mu = \mathbb{E}Z$, let ε be an independent Rademacher sign, and set

$$X = \varepsilon \sqrt{Z/\mu}.$$

Symmetry gives $\mathbb{E}X = 0$, while $\mathbb{E}X^2 = 1$. If $\mu t^2 \leq 1$, then $t < 1$ and $\mathbb{P}\{|X| \geq t\} \leq 1 \leq t^{-4}$. If $1 < \mu t^2 \leq L$, then

$$\mathbb{P}\{|X| \geq t\} = (\mu t^2)^{-2} \leq t^{-4},$$

and if $\mu t^2 > L$ the probability is zero. Hence X satisfies $\mathbf{H}(t^4)$ with parameter one.

Tail integration yields

$$\mu = 2 - L^{-1}, \quad \mathbb{E}Z^2 = 1 + 2 \log L = 1 + \log N, \quad \mathbb{E}Z^4 = 2N - 1.$$

Let $W = Z - \mu$ and $\sigma^2 = \mathbb{E}W^2$. Since $\mu \leq 2$,

$$\sigma^2 = 1 + \log N - \mu^2 \geq \log N - 3 \geq \frac{1}{2} \log N.$$

In particular $\sigma^4 \geq 9$. Also,

$$\mathbb{E}W^4 \leq 8(\mathbb{E}Z^4 + \mu^4) \leq 16N + 120 \leq 136N.$$

For independent copies $W_1, \dots, W_N$ and $R = \sum_i W_i$, independence and centering give

$$\mathbb{E}R^2 = N\sigma^2, \quad \mathbb{E}R^4 = N\mathbb{E}W^4 + 3N(N-1)\sigma^4 \leq 19(\mathbb{E}R^2)^2.$$

Indeed, the first term is at most $136N^2 < 16N^2\sigma^4$, and the second is at most $3N^2\sigma^4$. Paley–Zygmund applied to R^2 at level $1/2$ therefore gives [2]

$$\mathbb{P}\left\{|R| \geq \sqrt{N\sigma^2/2}\right\} \geq \frac{(1-1/2)^2(\mathbb{E}R^2)^2}{\mathbb{E}R^4} \geq \frac{1}{76}.$$

On this event, using $\mu \leq 2$,

$$\left| \frac{1}{N} \sum_{i=1}^N (X_i^2 - 1) \right| = \frac{|R|}{\mu N} \geq \frac{1}{4} \sqrt{\frac{\log N}{N}}.$$

Finally $Z \leq L$ and $\mu \geq 1$, so $N^{-1} \max_i X_i^2 \leq L/N = N^{-1/2}$ deterministically. $\square$

Corollary 2. *There is no universal constant C for which (1) holds under $\mathbf{H}(t^4)$ with success probability tending to one. Even the corresponding expectation bound fails by a factor of order $\sqrt{\log N}$.*

Proof. For $n = 1$, the proposed right side is at most $2C/\sqrt{N}$. If $\log N > 64C^2$, Theorem 1 violates it with probability at least $1/76$. The same event also gives

$$\mathbb{E} \left| \frac{1}{N} \sum_{i=1}^N (X_i^2 - 1) \right| \geq \frac{1}{304} \sqrt{\frac{\log N}{N}}.$$

$\square$

4 Verification, scope, and novelty

The example is full-strength with respect to dimension, centering, isotropy, and tail parameter. Its law depends on N , which is legitimate against a universal nonasymptotic estimate; within each row the variables are identically distributed and independent.

The result does *not* refute a slower endpoint rate or the stronger problem with a uniformly bounded fourth moment. In fact, the example has $\mathbb{E}X^4 \asymp \log N$, exactly exposing the distinction between weak L^4 and strong L^4 .

A bounded search covered the run indexes and exact combinations of “weak L^4 ”, “sample covariance”, “ $p = 4$ ”, and square-root/logarithmic rates. Tikhomirov proves a quantitative Bai–Yin theorem for finite $p > 4$ moments [3]. Abdalla and Zhivotovskiy prove a dimension-free finite- $p > 4$ version and still record a finite- $p = 4$ variant as open [4]. No source stating the present weak-tail endpoint counterexample was found. This is a bounded search, not an exhaustive novelty guarantee.

Human-review recommendation. The algebraic and probabilistic verification is short. Review should focus on the scope match: this packet completely disproves the square-root endpoint formulation, but deliberately does not label every possible interpretation of the source’s broader $p = 4$ sentence as solved.

References

- [1] O. Guedon, A. E. Litvak, A. Pajor, and N. Tomczak-Jaegermann, *On the interval of fluctuation of the singular values of random matrices*, arXiv:1509.02322 (2015).
- [2] R. E. A. C. Paley and A. Zygmund, *A note on analytic functions in the unit circle*, Proceedings of the Cambridge Philosophical Society 28 (1932), 266–272.
- [3] K. Tikhomirov, *Sample covariance matrices of heavy-tailed distributions*, arXiv:1606.03557 (2016).
- [4] P. Abdalla and N. Zhivotovskiy, *Covariance estimation: optimal dimension-free guarantees for adversarial corruption and heavy tails*, arXiv:2205.08494 (2023 revision).